

SUYASH SHARMA

B.E. Computer Science Engineering (AI & Machine Learning) | Chandigarh University

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TECHNICAL COMPETENCIES

Languages & Core: Python, C++, Data Structures and Algorithms, OOPs (def, class, inheritance), SQL

Machine Learning: Scikit-Learn, Supervised Learning, Random Forest Regressor, Cross-Validation, RMSE, Accuracy Metrics, Pandas, NumPy

Deep Learning & Vision: Deep Learning, Neural Networks, Computer Vision, PyTorch, Custom DataLoader, Loss Optimization, Epochs

Backend & Microservices: FastAPI, Uvicorn Server, Async Def Endpoints, Pydantic Schema Validation, PostgreSQL, Swagger UI

KEY PRODUCTION & MACHINE LEARNING PROJECTS

Air Pollution Forecasting & Atmospheric Intelligence Model

- Engineered a Machine Learning regression pipeline using **Scikit-Learn** and **Random Forest** on PM2.5 air quality datasets.
- Handled extensive data cleaning using **Pandas** and **NumPy**, performing 5-fold **cross-validation** to prevent overfitting.
- Optimized hyperparameter configurations to achieve minimized **RMSE** loss and 91.4% predictive **accuracy** metrics.

TrustMark AI — Real-time Identity Verification & Risk Engine

- Developed asynchronous microservice architecture powered by **FastAPI** and executed on high-performance **Uvicorn**.
- Implemented **Pydantic** models for request/response serialization with **async def** endpoints handling live queries.
- Integrated Computer Vision facial landmark models and Deep Learning scoring connected to **PostgreSQL** relational database.

Deep Learning Neural Classifier Lab

- Built Deep Learning vision models using **PyTorch**; trained neural networks tracking **epochs**, **loss**, and **Adam** optimizer.
- Implemented custom PyTorch **DataLoader** pipelines for tensor preprocessing and batch validation.

EDUCATION & ACADEMIC ACHIEVEMENTS

Bachelor of Engineering - CSE (AI & Machine Learning) — Chandigarh University (Current CGPA: 6.6)

- AI Fusion Program** — IIT Ropar (Advanced AI architectures & applied intelligence)
- Microsoft Elevate Program** — Engineering paradigms and cloud ecosystem certification